more rapidly than the upper strata of air. The result is a nearly continuous layer of stratus clouds about 1200 feet thick, lying at elevations varying between about 1000 and 3000 feet, that prevents air near the ground from being warmed. Temperatures, consequently, are relatively low. The coast of Peru thus is the cloudiest-and one of the driest-deserts in the world, with no sunshine for at least six months of the year. It almost never rains, except under abnormal circumstances, but condensation of fog provides a limited amount of moisture. A warm coastal desert extends from northeastern Colombia to Venezuela, covering a zone where rainfall is scarce and droughts are prolonged.

55. Caatinga (white forest) refers to the generally stunted, somewhat sparse, and often thorny vegetation of the dry interior of north-eastern Brazil. Trees, leafless for long periods and able to resist drought, also are a characteristic, particularly in the basin of the Sao Francisco River. Dominant species are leguminous trees.
56. Mountain vegetation: The upper zones have a peculiar vegetation that touches the snow line. In the wet northern Andes in Colombia and Ecuador, Alpine meadows called *páramo* consist of grasses and other herbaceous plants, often with bright flowers; those are surmounted by taller plants, especially the showy frailejones *(Espeletia),* which grow 15 to 20 feet (4.5 to 6 metres) high and are crowned by large bouquets of long hairy leaves.

    In the south, *páramo* vegetation merges with that of the high plateaus, notably the *tola*, a tough wind- resistant shrub, in the Altiplano of southeastern Peru and western Bolivia. Typical vegetation consists of rough grasses, between which grow a variety of herbaceous, cushion, and rosette plants, shrubs, and cacti. Vegetation extends to the limit of permanent snow but becomes scarce at higher elevations, where soil is often barren.
57. The line passes through Chile, Argentina, Paraguay, Brazil.
58. Paraguay borders on three substantially larger countries: Bolivia, Brazil, and Argentina.
59. The Amazon River in South America is the largest river by discharge volume of water in the world and discharges into Atlantic ocean.

    Spreads over Brazil, Colombia, Peru.